

Zomato Data Analysis Project

In [2]: *# Step 1 - Importing Libraries*

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

In [3]: *# step 2 - creating DataFrame*

```
df = pd.read_csv("Zomato data .csv")

df
```

Out[3]:

	name	online_order	book_table	rate	votes	approx_cost(for two people)	listed_
0	Jalsa	Yes	Yes	4.1/5	775	800	
1	Spice Elephant	Yes	No	4.1/5	787	800	
2	San Churro Cafe	Yes	No	3.8/5	918	800	
3	Addhuri Udupi Bhojana	No	No	3.7/5	88	300	
4	Grand Village	No	No	3.8/5	166	600	
...	...	...	...	...	...	...	...
143	Melting Melodies	No	No	3.3/5	0	100	
144	New Indraprasta	No	No	3.3/5	0	150	
145	Anna Kuteera	Yes	No	4.0/5	771	450	
146	Darbar	No	No	3.0/5	98	800	
147	Vijayalakshmi	Yes	No	3.9/5	47	200	

148 rows × 7 columns

In [4]: *# convert the datatype of column - rate*

```
def handleRate(value):
    value = str(value).split('/')
    value = value[0];
    return float(value)

df['rate'] = df['rate'].apply(handleRate)

df.head()
```

```
Out[4]:
```

	name	online_order	book_table	rate	votes	approx_cost(for two people)	listed_in(type)
0	Jalsa	Yes	Yes	4.1	775	800	
1	Spice Elephant	Yes	No	4.1	787	800	
2	San Churro Cafe	Yes	No	3.8	918	800	
3	Addhuri Udupi Bhojana	No	No	3.7	88	300	
4	Grand Village	No	No	3.8	166	600	

```
In [5]: df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 148 entries, 0 to 147
Data columns (total 7 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   name                                  148 non-null    str
1   online_order                         148 non-null    str
2   book_table                           148 non-null    str
3   rate                                 148 non-null    float64
4   votes                                148 non-null    int64
5   approx_cost(for two people)          148 non-null    int64
6   listed_in(type)                       148 non-null    str
dtypes: float64(1), int64(2), str(4)
memory usage: 8.2 KB
```

```
In [6]: df.isnull().sum()
```

```
Out[6]: name                                0
online_order                              0
book_table                                0
rate                                      0
votes                                    0
approx_cost(for two people)              0
listed_in(type)                          0
dtype: int64
```

```
In [7]: df.shape
```

```
Out[7]: (148, 7)
```

```
In [8]: df.describe()
```

```
Out[8]:
```

	rate	votes	approx_cost(for two people)
count	148.000000	148.000000	148.000000
mean	3.633108	264.810811	418.243243
std	0.402271	653.676951	223.085098
min	2.600000	0.000000	100.000000
25%	3.300000	6.750000	200.000000
50%	3.700000	43.500000	400.000000
75%	3.900000	221.750000	600.000000
max	4.600000	4884.000000	950.000000

```
In [9]: # Type of resturant majority customers order from
```

```
df.head()
```

```
Out[9]:
```

	name	online_order	book_table	rate	votes	approx_cost(for two people)	listed_in(type)
0	Jalsa	Yes	Yes	4.1	775	800	
1	Spice Elephant	Yes	No	4.1	787	800	
2	San Churro Cafe	Yes	No	3.8	918	800	
3	Addhuri Udupi Bhojana	No	No	3.7	88	300	
4	Grand Village	No	No	3.8	166	600	

```
In [ ]: sns.countplot(data= df , x=df['listed_in(type)'], palette='Set2')

plt.xlabel('Type of Resturants')

plt.title('Zomato - Resturants')

plt.show()
```

```
In [ ]: #conclusion - Majoity of the returant falls in 'Dinning' category
```

```
In [ ]: df.head()
```

```
Out[ ]:
```

	name	online_order	book_table	rate	votes	approx_cost(for two people)	listed_in(type)
0	Jalsa	Yes	Yes	4.1	775	800	
1	Spice Elephant	Yes	No	4.1	787	800	
2	San Churro Cafe	Yes	No	3.8	918	800	
3	Addhuri Udupi Bhojana	No	No	3.7	88	300	
4	Grand Village	No	No	3.8	166	600	

```
In [ ]: grouped_data = df.groupby('listed_in(type))['votes'].sum()

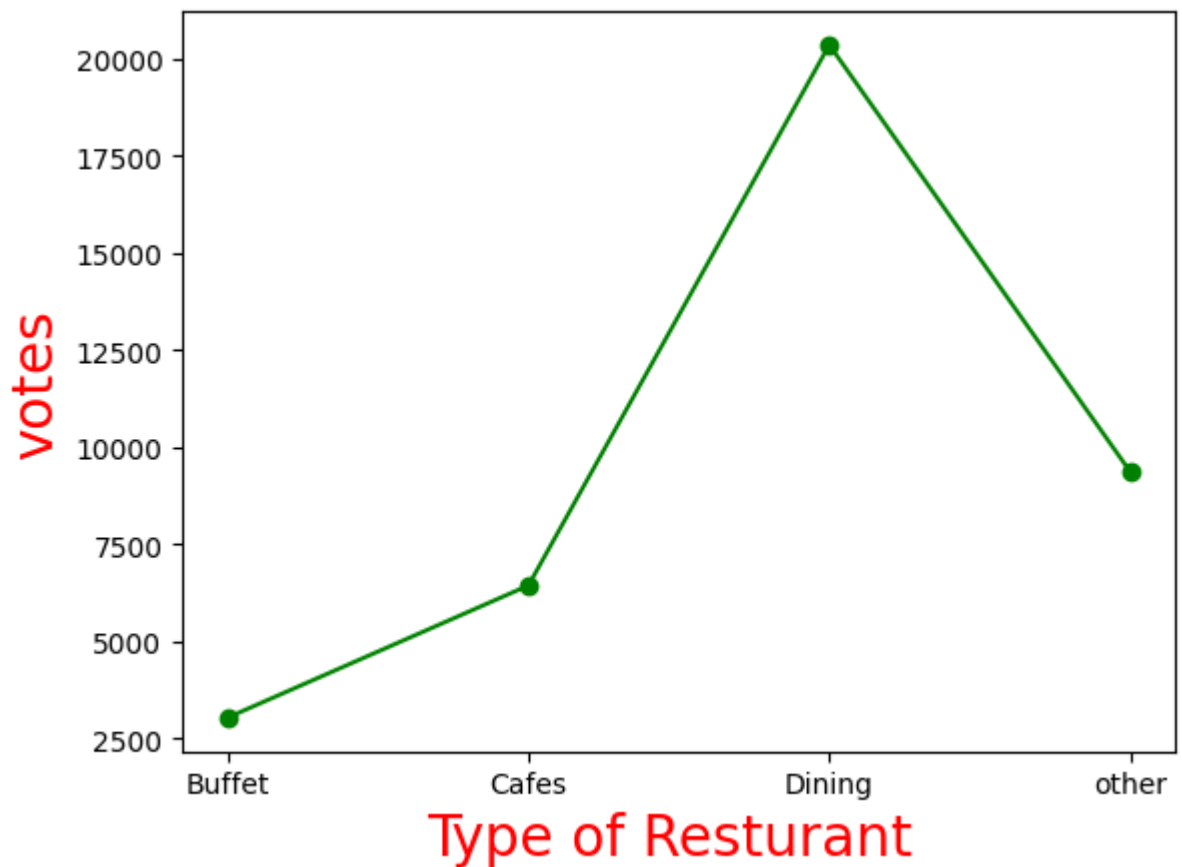
result = pd.DataFrame({'vote': grouped_data})

plt.plot(result,c="green",marker='o')

plt.xlabel("Type of Resturant ", c="red" ,size=20)

plt.ylabel("votes",c="red",size=20)
```

```
Out[ ]: Text(0, 0.5, 'votes')
```



```
In [ ]: # conclusion: Dinning resturants has recieved maximum votes
```

```
In [ ]: grouped_data
```

```
Out[ ]: listed_in(type)
Buffet    3028
Cafes     6434
Dining    20363
other     9367
Name: votes, dtype: int64
```

```
In [ ]: df.head()
```

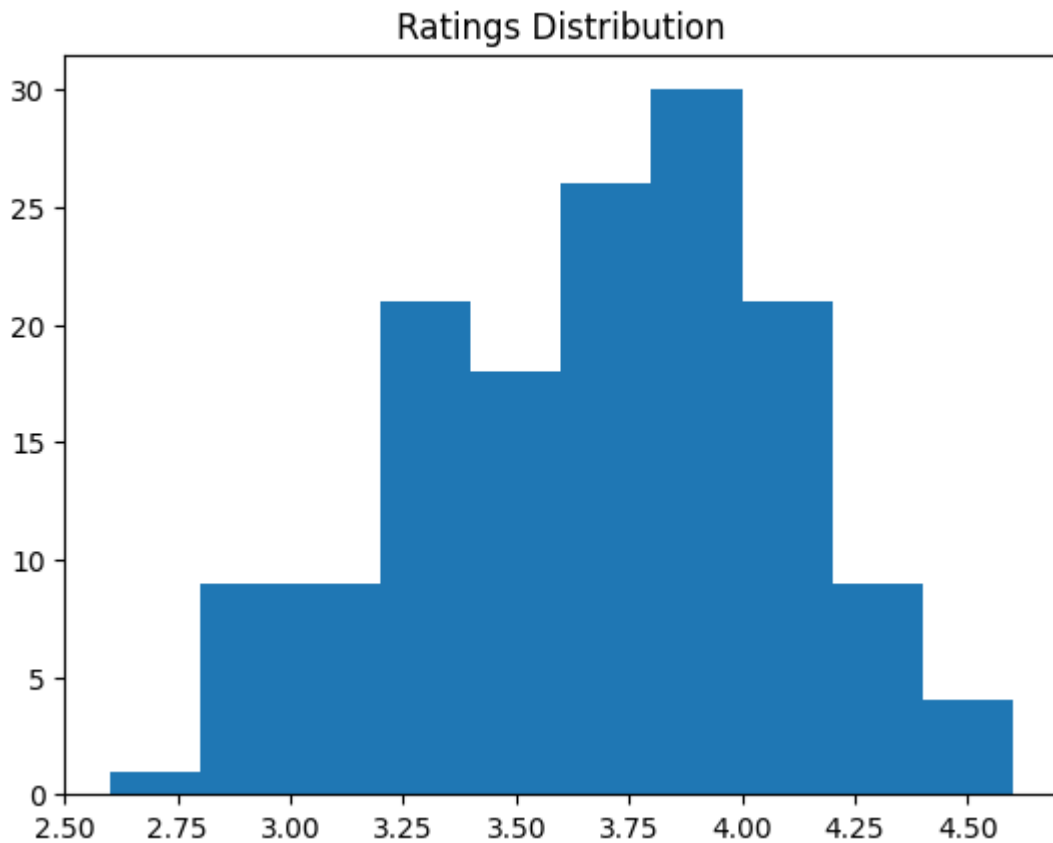
```
Out[ ]:
```

	name	online_order	book_table	rate	votes	approx_cost(for two people)	listed_
0	Jalsa	Yes	Yes	4.1	775	800	
1	Spice Elephant	Yes	No	4.1	787	800	
2	San Churro Cafe	Yes	No	3.8	918	800	
3	Addhuri Udupi Bhojana	No	No	3.7	88	300	
4	Grand Village	No	No	3.8	166	600	

```
In [ ]: plt.hist(df['rate'],bins=10)

plt.title('Ratings Distribution')

plt.show()
```



```
In [ ]: # conclusion: the majority resturants recieved ratings from 3.5 to 4
```

```
In [ ]: # Average Order Speding by Couples
```

```
df.head()
```

```
Out [ ]:
```

	name	online_order	book_table	rate	votes	approx_cost(for two people)	listed_
0	Jalsa	Yes	Yes	4.1	775	800	
1	Spice Elephant	Yes	No	4.1	787	800	
2	San Churro Cafe	Yes	No	3.8	918	800	
3	Addhuri Udupi Bhojana	No	No	3.7	88	300	
4	Grand Village	No	No	3.8	166	600	

```
In [ ]: couple_data = df['approx_cost(for two people)']

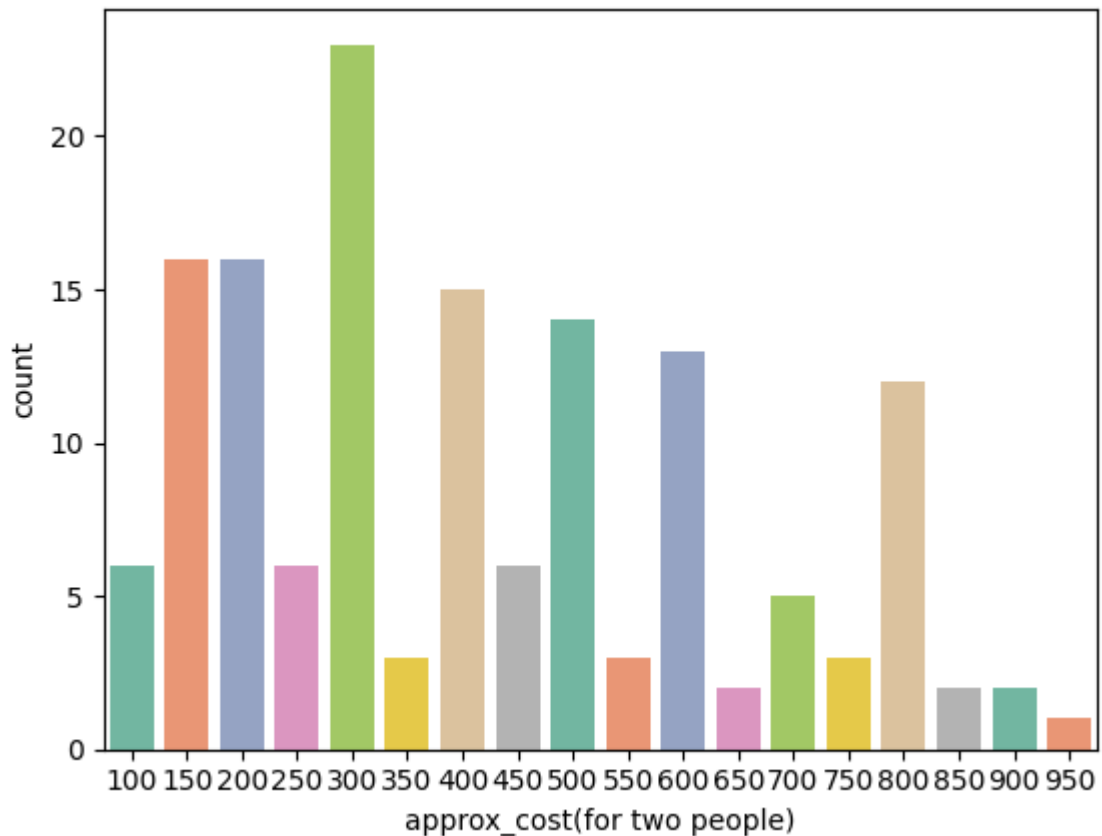
sns.countplot(x=couple_data , palette='Set2')
```

C:\Users\rutuu\AppData\Local\Temp\ipykernel_3812\4059947118.py:3: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14. Assign the `x` variable to `hue` and set `legend=False` for the same effect.

```
sns.countplot(x=couple_data , palette='Set2')
```

```
Out [ ]: <Axes: xlabel='approx_cost(for two people)', ylabel='count'>
```



```
In [ ]: # conclusion : the mojority of couples prefer returants with an approximate cost
```

```
In [ ]: # Which mode receives max ratings
```

```
df.head()
```

```
Out [ ]:
```

	name	online_order	book_table	rate	votes	approx_cost(for two people)	listed_
0	Jalsa	Yes	Yes	4.1	775	800	
1	Spice Elephant	Yes	No	4.1	787	800	
2	San Churro Cafe	Yes	No	3.8	918	800	
3	Addhuri Udupi Bhojana	No	No	3.7	88	300	
4	Grand Village	No	No	3.8	166	600	

```
In [ ]: plt.figure(figsize=(6,5))

sns.boxplot(data= df, x='online_order', y='rate' , palette='Set2')

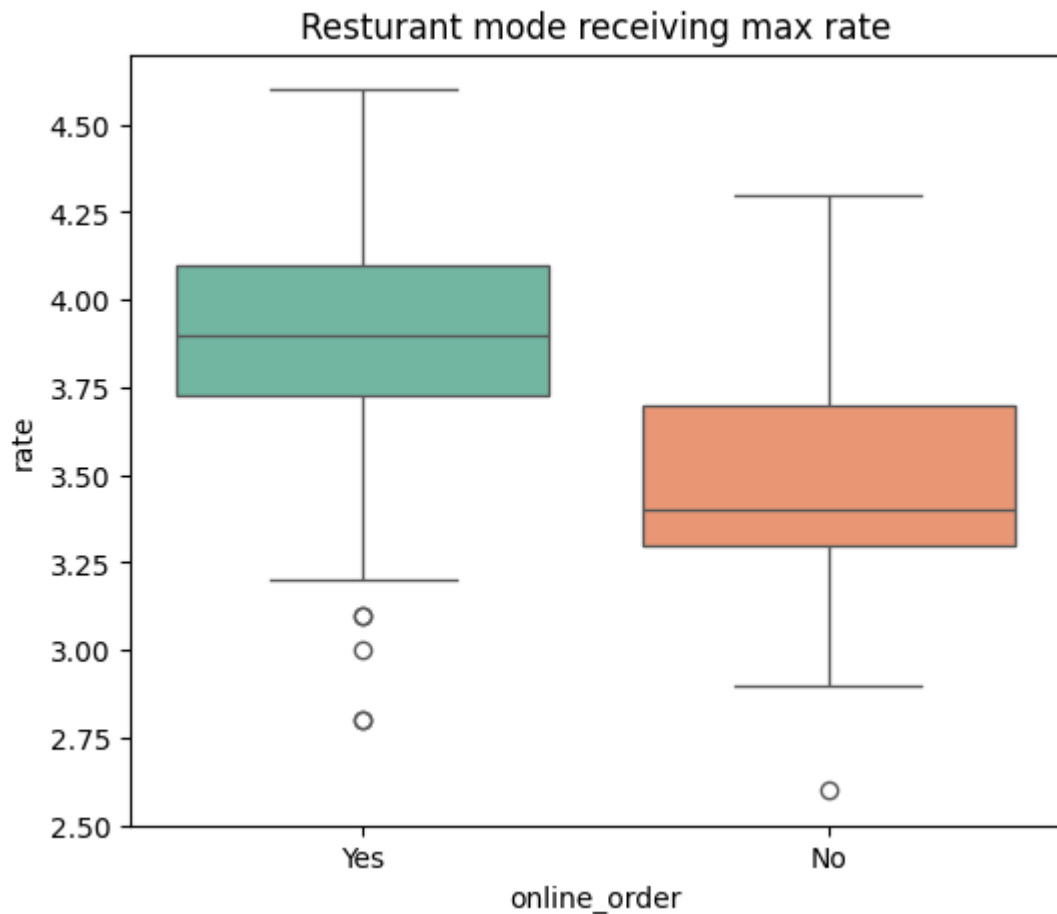
plt.title('Resturant mode receiving max rate')
```

C:\Users\rutuu\AppData\Local\Temp\ipykernel_3812\2968609547.py:3: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14. Assign the `x` variable to `hue` and set `legend=False` for the same effect.

```
sns.boxplot(data= df, x='online_order', y='rate' , palette='Set2')
```

```
Out [ ]: Text(0.5, 1.0, 'Resturant mode receiving max rate')
```



In []: *# conclusion : offline order received lower rating in comparison to online order*

In []: `df.head()`

Out[]:

	name	online_order	book_table	rate	votes	approx_cost(for two people)	listed_in
0	Jalsa	Yes	Yes	4.1	775	800	
1	Spice Elephant	Yes	No	4.1	787	800	
2	San Churro Cafe	Yes	No	3.8	918	800	
3	Addhuri Udupi Bhojana	No	No	3.7	88	300	
4	Grand Village	No	No	3.8	166	600	

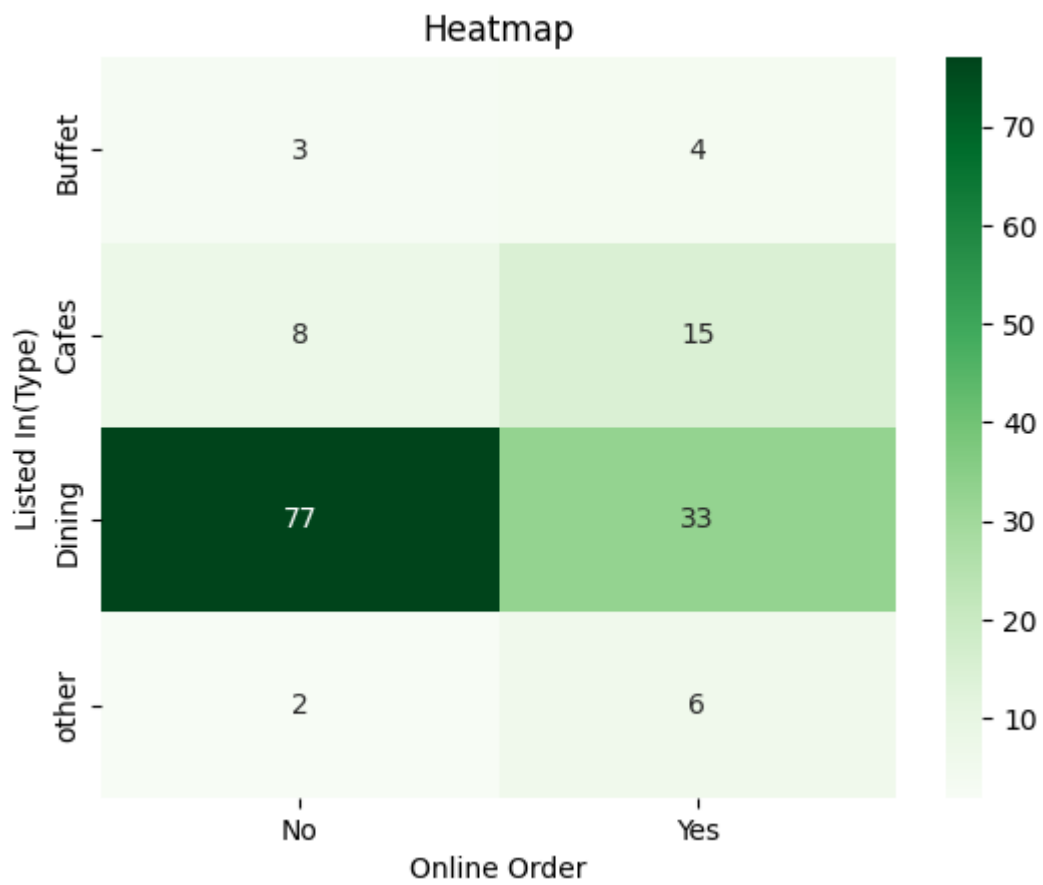
```
In [ ]: pivot_table = df.pivot_table(index='listed_in(type)', columns = 'online_order',
sns.heatmap(pivot_table, annot=True, cmap='Greens' , fmt='d')

plt.title("Heatmap")

plt.xlabel("Online Order")

plt.ylabel("Listed In(Type)")

plt.show()
```



```
In [ ]: # Conlcusion : Dinning type of Resturants received more offline orders so that 2
```